Anurag Lnu

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PROFESSIONAL SUMMARY

- **AI Engineer / ML Engineer** with **5 years** building high-scale **distributed systems** and production services; currently pursuing an MS in CS focused on **RAG** and **Computer Vision**.
- Specialize in **production ML systems**: data/embedding pipelines, retrieval + generation, model evaluation, and **inference APIs** with performance, reliability, and observability.
- Built end-to-end AI products including **TigerResearchBuddy** (local-first RAG with vector search + grounded answers) and ML systems with **FastAPI** serving + dashboards.

SKILLS

AI/ML: Python, RAG, MCP, NLP, PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers, LLMs, Embeddings, Vector Search, Computer Vision, Fine-tuning, Representation Learning, Model Evaluation, Error Analysis
MLOps & Serving: Docker, Model Packaging, FastAPI, REST, CI/CD, Monitoring/Observability, A/B testing, Offline/Online evaluation, Latency optimization
Data & Systems: PySpark, Pandas, Feature Engineering, ETL, Kafka, Redis, Microservices, AsyncIO
Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Vector DB (ChromaDB)
Languages: Python, Java, Scala, JavaScript, TypeScript, C
Tools: Git, GitHub, Jenkins, OpenShift, Prometheus, Grafana, SonarQube

PROJECTS

TigerResearchBuddy — Local-First RAG Research Assistant RAG / Vector DB / Local LLM *Local-first RAG assistant: scrape + PDF ingest, embed + index in ChromaDB, answer with Ollama; Streamlit UI.*

- Built a private, local-first system combining **web scraping + PDF ingestion** with **RAG** and **local LLM inference** (Ollama) for grounded Q&A.
- Implemented an embedding pipeline (chunking → embeddings → indexing) using **ChromaDB** for fast **semantic search** and retrieval.
- Shipped an interactive **Streamlit** UI and batch refresh workflows to support repeatable corpus updates and offline usage.
- **Tech:** Python, Streamlit, Ollama, ChromaDB **ML/Data:** Embeddings, vector search, grounded generation

Hybrid ML Scheduler — Real-Time Heterogeneous Scheduling Lab ML Systems / RL / Dashboard *Heterogeneous scheduling benchmark with 6 policies, online retraining, and live metrics dashboard (FastAPI + React + WebSockets).*

- Built a simulation framework comparing **6 scheduling strategies** including heuristic and learning-based policies for heterogeneous environments.
- Implemented a live telemetry pipeline via **WebSockets** with **FastAPI** backend + **React** dashboard to stream metrics and visualize performance.
- Added **online learning** (periodic retraining) and tracked **time/energy/cost** to enable benchmark-style evaluation and iteration.
- **Tech:** Python, FastAPI, WebSocket, React **ML/Data:** RL, online retraining, systems benchmarking

Smart Bets — ML-Driven Virtual Betting System End-to-End ML / Serving *End-to-end ML pipeline (XGBoost on 16K+ matches) with edge/confidence filters, Kelly sizing, and FastAPI + React dashboard.*

- Trained an **XGBoost** model on **16,000+** matches and built a full pipeline: feature extraction → prediction → filtering → execution.
- Implemented **edge/confidence gates** and **Kelly Criterion** sizing with bankroll controls; automated scheduled runs and persisted outcomes for evaluation.
- Exposed predictions + metrics via **FastAPI** and a **React** dashboard to keep decisions inspectable and iteration-friendly.
- **Tech:** Python, XGBoost, FastAPI, React, SQLite **ML/Data:** Backtesting, risk controls, evaluation loop

dART — Distributed Adaptive Resonance Theory (ART-1) Distributed Neural Systems *Distributed ART-1 neural network in Scala/Akka actors with tests and a NumPy baseline.*

- Implemented **ART-1** using **Akka Actor Model** in **Scala** to study stability–plasticity via distributed, asynchronous message passing.
- Built core ART-1 components (F1/F2 layers + orienting subsystem) with vigilance-controlled category formation and stable memory.
- Added demos/tests and a baseline (**Python/NumPy**) to validate correctness and compare behavior.

- **Tech:** Scala, Akka, SBT, ScalaTest **ML/Data:** Actor-style message passing, distributed simulation

NYPD 911 Calls to CrimeCastNYC 2.0 — Advanced Crime Analytics Platform Big Data / Analytics *PySpark ETL + PostgreSQL star schema to mine spatiotemporal patterns and association rules.*

- Built a scalable **ETL** pipeline: ingestion (Socrata API), **PySpark** cleaning into Parquet, and bulk load into a **star schema** in **PostgreSQL**.
- Engineered mining-ready representations: normalized complaint codes into semantic groups and generated transactional itemsets for stable pattern discovery.
- Implemented **Apriori** association rules (lift/confidence) and baseline spatiotemporal modeling for forecasting/triage insights.
- **Tech:** PySpark, Python, PostgreSQL, MongoDB, Docker **ML/Data:** Star schema, partitioning/indexing, itemsets, Apriori

Image Similarity Engine Computer Vision / Retrieval *PyTorch embedding model for image similarity with kNN retrieval and CPU inference pipeline.*

- Built an embedding-based similarity system: learned **visual embeddings** and served fast **kNN** retrieval for image search.
- Fine-tuned a compact CNN embedding model and shipped a low-latency CPU inference pipeline with checkpoint versioning and model selection.
- **Tech:** PyTorch, OpenCV **ML/Data:** Embeddings, nearest-neighbor retrieval, inference pipeline

EXPERIENCE

Ernst & Young (EY), Bengaluru, India **Apr 2022 – Jul 2024**
Senior Technology Consultant

- Built and operated high-throughput backend services generating **100,000+** monthly client artifacts; improved correctness via streaming validation and automated quality checks.
- Designed event-driven data pipelines using **Kafka** to reduce reconciliation effort by **30%** and improve data accuracy by **15%**.
- Implemented distributed caching/queueing patterns with **Redis** to optimize hot paths, reducing average response times by **70%** under production traffic.
- Delivered profiling-driven performance improvements that reduced pipeline latency by **25%** through measurement + iterative optimization.
- Built real-time decisioning workflows for fraud monitoring, reducing review time by **75%** with reliable event processing and automated triage queues.
- **Tech:** Java, Scala, Spring Boot, Kafka, Redis, SQL **ML/Data:** Event-driven systems, performance tuning, reliability patterns

Tata Consultancy Services (TCS), Hyderabad, India **Jul 2019 – Apr 2022**
Software Engineer

- Engineered resilient APIs processing **1M+** daily transactions with **99.9%** uptime using production reliability patterns and monitoring.
- Integrated **Kafka** event streams and **Cassandra** clusters, reducing data retrieval latency by **20%** for high-concurrency workloads.
- Established observability with **Prometheus** and **Grafana**, cutting MTTR by **35%** through faster detection and diagnosis.
- Built CI/CD pipelines and containerized workloads with **Docker**/OpenShift, reducing deployment lead time by **40%** and improving release consistency.
- Optimized relational and NoSQL schemas (**MySQL**, **MongoDB**, **Cassandra**), reducing query times by **50%** via indexing and data-model tuning.
- **Tech:** Spring Boot, Kafka, Cassandra, Docker/OpenShift, Jenkins **ML/Data:** APIs, observability, CI/CD, schema tuning

EDUCATION

- **Rochester Institute of Technology (RIT), Rochester, NY** **Aug 2024 – Present**
  Master of Science in Computer Science *Focus: AI, RAG, Computer Vision*
- **Sikkim Manipal Institute of Technology, India** **Jun 2015 – Jul 2019**
  B.Tech in Computer Science & Engineering

CERTIFICATIONS

- Oracle Certified Associate, Java SE 8 Programmer
- AWS Cloud Practitioner